

MICROBIOLOGY – SHORT EXAM NOTES

Biomedical Waste Management • Introduction to HAI • Prevention & Control – Standard Precautions
(Ananthanarayan & Paniker, Ch. 69–70)

1. BIOMEDICAL WASTE MANAGEMENT

Definition (BMW Rules, Govt. of India 2016): any waste generated during the **diagnosis, treatment or immunisation** of human beings or animals, or in research activities pertaining thereto, or in the production or testing of biologicals. The persons in charge of institutions generating such waste ('**occupiers**') are legally responsible for its safe management.

Why Biomedical Waste is Hazardous

- Contains **infectious and hazardous** material that may harm patients, visitors, staff and the public.
- Contains **sharps** (needles, broken glass) causing injury and infection, and harmful chemicals or radioactive material.
- Untreated waste ferments and attracts **flies, insects, birds and animals**; attracts **rag pickers** who repack and resell disposables and drugs.
- Pollutes **soil, water and air**.

Types of Biomedical Waste

In India about **1–2 kg of waste is generated per bed per day**, of which about **85% is harmless** and only **15% is hazardous**.

Type	Examples
Harmless (general) waste	Paper, cardboard, cartons, flowers, office and kitchen waste
Infectious waste	Human/animal tissues and organs, placenta, pathological fluids, soiled dressings and swabs, laboratory samples
Non-infectious hazardous waste	Chemical (toxic, corrosive, inflammable), radioactive (under Bhabha Atomic Research Centre), pharmaceutical (expired drugs)

Steps in Waste Management

1. **Reduction** – minimise volume; use reusable items where safe.
2. **Segregation** – separation **at the point of generation** into colour-coded containers. The **most essential step**: mixing harmful with harmless waste contaminates the whole load.
3. **Storage** – safe keeping at the point of generation until collection.
4. **Transportation** – movement from the point of generation to the treatment facility.
5. **Treatment and disposal** – to reduce bulk and make waste non-infectious and harmless.

Colour Coding and Treatment (BMW Rules 2016)

Colour	Waste category	Treatment / disposal
YELLOW	Human and animal anatomical waste, soiled waste (dressings, plaster casts), expired/discarded medicines, chemical waste, discarded linen soiled with blood, microbiology and lab waste (pre-treated)	Incineration , plasma pyrolysis or deep burial ; lab waste pre-treated on site (autoclave/chemical) before incineration; expired drugs returned to manufacturer

Colour	Waste category	Treatment / disposal
RED	Contaminated recyclable waste – tubing, bottles, IV sets, catheters, urine bags, syringes without needles, gloves	Autoclaving/microwaving/hydroclaving , then shredding and sent to registered recyclers
WHITE (translucent, puncture-proof)	Sharps – needles, syringes with fixed needles, scalpels, blades	Autoclaving or dry heat, then shredding/mutilation or encapsulation; sharps pit / iron foundry
BLUE (cardboard box with blue label)	Glassware – broken/contaminated glass, medicine vials and ampoules, metallic body implants	Disinfection (hypochlorite) or autoclaving, then recycling

Methods of Treatment

- **Chemical disinfection** – e.g. hypochlorite; useful in small clinics and as pre-treatment (sputum, pus).
- **Incineration** – very high temperature; converts waste to ash (~1/10 volume); for **anatomical waste, amputated limbs, carcasses**. Expensive.
- **Deep burial** – after disinfection, placed in deep trenches, covered with **lime** and soil; also for sharps where land is available.
- **Autoclaving** – widely used in laboratories for infectious waste before disposal.
- **Microwaving** – small volumes at the point of generation (not for body parts, metals, toxic or radioactive waste).
- **Inertisation** – mixing waste with cement to stop toxic substances leaching into water.
- **Liquid waste** – disinfected/neutralised before flushing into the sewer.

Proper waste disposal is an important measure to prevent HAI – '**Primum non nocere**' – **First, do no harm**.

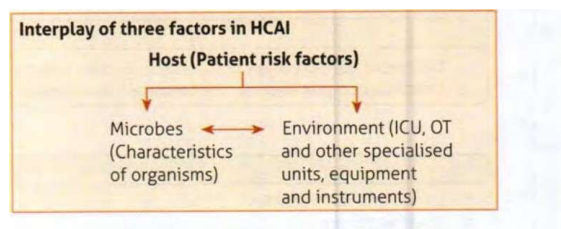
★ EXAM FOCUS

- Short notes: **Types of biomedical waste, Segregation / colour coding of BMW, BMW management in hospitals**.
- Key numbers: 1–2 kg/bed/day; 15% hazardous; BMW Rules **2016**; segregation at source is the most important step.

2. INTRODUCTION TO HEALTHCARE-ASSOCIATED INFECTIONS (HAI)

Healthcare-associated infections (HCAI/HAI), earlier called **hospital-acquired or nosocomial infections**, are infections acquired in a healthcare facility that were **neither present nor incubating at the time of admission**. They usually appear **48–72 hours after admission**, and infections that appear **after discharge** are also included. The term 'healthcare-associated' is used because they can occur in any healthcare facility, not only hospitals. **Iatrogenic infection** (resulting from treatment or investigations) is a related but separate concept.

Magnitude: HAI affects **5–10%** of admitted patients in developed countries and **more than 25%** in developing countries, increasing morbidity, mortality and cost. Up to **20% (developed)** and **40% (developing)** of HAIs are preventable.



Interplay of three factors in HCAI – host, microbes and environment (from the book)

Common Types of HAI

Infection	Key points	Common organisms
Catheter-associated UTI (CAUTI)	Most common; ~80% linked to indwelling catheters; 2% after single catheterisation, ≥50% with indwelling catheters. Prevent: closed sterile drainage, short duration, 'bundle' care	<i>E. coli</i> , <i>Proteus</i> , <i>E. faecalis</i> , <i>Ps. aeruginosa</i>
Bloodstream infection (BSI)	Mainly from IV cannulae (central/peripheral lines); risk rises with duration; change cannula every 48 h	<i>S. epidermidis</i> , <i>S. aureus</i> , <i>E. faecalis</i> , <i>Candida</i>
Pneumonia / VAP	Aspiration, ventilators, instrumentation; MDR <i>Acinetobacter baumannii</i> rising	<i>Klebsiella</i> , <i>Pseudomonas</i> , <i>S. aureus</i> (MRSA)
Surgical site infection (SSI)	Depends on site/duration of surgery, patient, operator; appear within a week (<i>S. pyogenes</i> 1–2 days, staphylococci 4–5 days, Gram-negatives 6–7 days)	<i>S. aureus</i> , <i>E. coli</i> , <i>E. faecalis</i>
Transfusion-associated	Hepatitis B and C in dialysis, transfusion; risk to staff handling blood	HBV, HCV, HIV
Gastroenteritis	Food poisoning outbreaks, antibiotic-associated diarrhoea	<i>Salmonella</i> , <i>S. aureus</i> , <i>C. difficile</i>

Sources and Reservoirs

- **Endogenous** – the patient's own flora; often the **most common** source, especially with debilitating disease, invasive procedures, immunosuppressants or antibiotics.
- **Cross-infection** – from other patients or **hospital personnel** by direct contact (hands, droplets), air or carriage on staff.
- **Environmental** – hospital air, contaminated surfaces, and inanimate objects (beds, sanitary fittings, endoscopes, catheters, needles, instruments, kitchen, laundry).

Modes of Transmission

Four main routes: **aerial, oral, contact (especially 'hand-borne' – the most important)** and **parenteral**.

3. PREVENTION AND CONTROL OF HAI – STANDARD PRECAUTIONS

Infection control aims to: (1) **reduce the microbial population** of the hospital environment; (2) **prevent transmission** – staff to patient, patient to staff to patient, patient to patient; (3) manage linen, equipment and objects so they do not become sources of cross-infection; and (4) practise **safe biomedical waste disposal**.

Standard precautions are the **minimum infection prevention practices that apply to all patient care, regardless of the suspected or confirmed infection status of the patient**, in any healthcare setting. The principle: **consider every person as potentially infectious and susceptible to infection**.



Fig. 69.1 Standard precautions (from the book)

1. Hand Hygiene

The **single most important** measure. Done by **alcohol-based hand rub** or **handwashing with soap and water** (15–20 seconds, 7 steps: palm to palm; right palm over left dorsum with interlaced fingers and vice versa; palm to palm fingers interlaced; backs of fingers to opposing palms; rotational rubbing of thumbs; rotational rubbing of fingertips in palm; rotational rubbing of wrists), then rinse and dry with a single-use towel, turning off the tap with the same towel.



Fig. 69.2 The ideal handwashing technique (from the book)

Five moments for hand hygiene (WHO): 1. Before touching a patient; 2. Before a clean/aseptic procedure; 3. After body fluid exposure risk; 4. After touching a patient; 5. After touching patient surroundings. Hands are also cleaned after removing gloves.

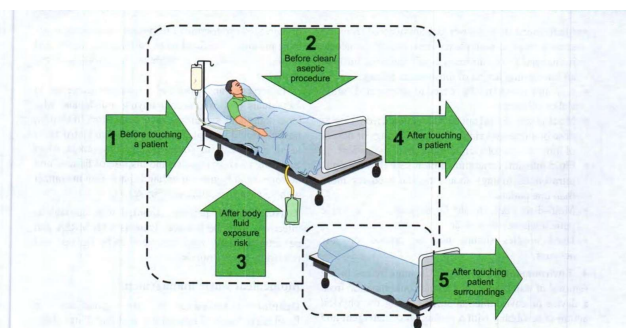


Fig. 69.3 Five moments for hand hygiene (from the book)

2. Personal Protective Equipment (PPE)

Wearable equipment protecting healthcare workers: **gloves, gowns, masks, respirators, goggles and face shields**.

- **Gloves** when contact with blood, body fluids, mucous membranes, non-intact skin or contaminated equipment is possible; **never reuse or wash** gloves; change between patients; hand hygiene after removal.
- **Gown** when splashes of blood/body fluid are expected; one gown per patient.
- **Mask, eye protection/face shield** for procedures generating splashes; surgical mask for spinal procedures.

3. Safe Injection Practices

- Aseptic technique; clean vial diaphragms with **70% alcohol**.
- **One needle, one syringe, one patient** – never use the same syringe for multiple patients, even if the needle is changed.
- Single-dose vials and IV sets for one patient only; multi-dose vials dedicated to one patient where possible.
- **Never recap used needles**; discard sharps into puncture-proof containers.

4. Environmental Cleaning, Linen and Equipment

Cleaning (removal of visible soil with detergent and water) must always precede **disinfection**. Reusable equipment (endoscopes, instruments) must be cleaned and disinfected/sterilised before use on another patient; single-use devices must not be reprocessed. Linen is handled so it does not spread contamination.

5. Respiratory Hygiene / Cough Etiquette

Identify patients with cough or respiratory symptoms at entry; cover the mouth and nose with a tissue or upper sleeve when coughing or sneezing, dispose of tissues, and wash hands afterwards.

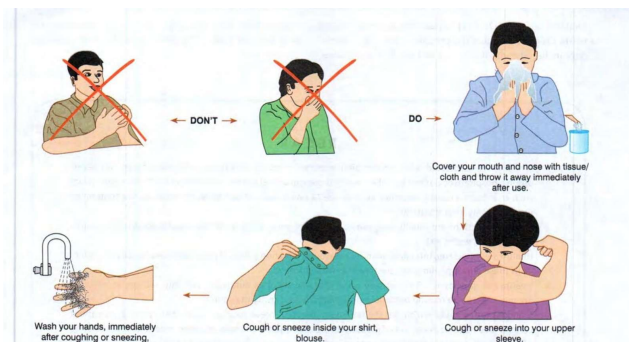


Fig. 69.4 Respiratory hygiene while coughing and sneezing (from the book)

6. Isolation and Waste Disposal

Patients with contagious infections are **isolated**; those with **MRSA or pan-drug-resistant organisms** need isolation and **barrier nursing**. Biomedical waste is segregated in colour-coded bags, sharps in puncture-proof containers, and infectious waste autoclaved before disposal.

Other Control Measures

- **Operation theatre:** conventional OTs have ~20 air changes/hour; **laminar flow OTs up to 300** (≤ 10 cfu/m³). Bioload should not exceed **35 in an empty theatre or 180 during an operation**.

- **Outbreak investigation:** look for carriers, a single source (e.g. food), faulty techniques; notify notifiable diseases.
- **Hospital Infection Control Committee (HICC):** chaired by the **Medical Superintendent**; **head of Microbiology is member secretary**; the **infection control nurse (ICN)** monitors wards, OTs, CSSD, laundry and waste units. It monitors infection rates and practices, detects outbreaks and trains staff.

★ EXAM FOCUS

- Short notes: **Types of HAI, Hand hygiene, PPE, Safe injection practices, Role of HICC, Standard precautions.**
- Define nosocomial infection – appears **48–72 h after admission**, not present or incubating at admission.
- Most common HAI – **UTI (CAUTI)**; most important preventive measure – **hand hygiene**.